

Statistics: Standard Deviation — Solutions

Full Step-by-Step | Trap Alerts | Key Insights | Fast Methods | Desmos Verdicts

Q1

Easy

Answer: B) SD of A < SD of B

Step-by-Step Solution:

Data set A = {3,3,3,3,3}: every value equals the mean (3), so all deviations are 0.

Standard deviation of A = 0.

Data set B = {1,2,3,4,5}: values vary around the mean (3).

Standard deviation of B > 0.

Therefore $SD(A) < SD(B)$.

Why Other Options Are Wrong:

A) $SD(A)$ is 0, which cannot be greater than any non-negative number.

C) $SD(A) = 0$, $SD(B) > 0$, so they are not equal.

D) We have full information — both data sets are given completely.

TRAP ALERT

Students might think identical values means "standard" or "normal" and assign a positive SD. A data set with zero variation has $SD = 0$, always.

KEY INSIGHT

$SD = 0$ if and only if every value in the data set is the same. This is a fundamental property tested repeatedly.

FAST METHOD

All identical values $\rightarrow SD = 0$. Any variation $\rightarrow SD > 0$. Compare instantly.

VERDICT

Hand-solve. Pure conceptual — 5 seconds.

Q2

Easy

Answer: B) Every value is 50

Step-by-Step Solution:

Standard deviation of 0 means zero spread — no value deviates from the mean.

If $SD = 0$, every data point must equal the mean.

Since mean = 50, every value = 50.

Why Other Options Are Wrong:

A) The data set could have any number of values (all equal to 50), not just one.

C) The number of values can be odd or even.

D) If all values are 50, then median = 50 = mean, so median is not greater.

TRAP ALERT

Option A is tempting but wrong — a set with 100 values of 50 also has $SD = 0$.

KEY INSIGHT

$SD = 0$ means every value equals the mean. This is the defining property.

FAST METHOD

$SD = 0 \Rightarrow$ all values equal $\Rightarrow$ all are 50.

VERDICT

Hand-solve. Conceptual — 5 seconds.

Q3

Easy

Answer: B) It decreases

Step-by-Step Solution:

Original ages: 28, 31, 35, 31, 40. Mean = $165/5 = 33$.

The value 40 is farthest from the mean (deviation = 7).

New ages: 28, 31, 35, 31, 32. Mean = $157/5 = 31.4$.

The new value (32) is closer to the new mean than 40 was to the old mean.

Replacing an extreme value with one closer to the center reduces spread, so SD decreases.

Why Other Options Are Wrong:

A) Replacing an outlier with a central value decreases spread.

C) The spread clearly changes when an extreme value is removed.

D) We have full information to determine the direction of change.

TRAP ALERT

Students sometimes think any change to the data makes the effect unpredictable. Here, we replaced the most extreme value with one near the center — SD must decrease.

KEY INSIGHT

Adding/replacing a value closer to the mean decreases SD. Adding/replacing one farther from the mean increases SD.

FAST METHOD

40 is the outlier. Replacing it with 32 (near the center) $\rightarrow$ SD decreases. No calculation needed.

VERDICT

Hand-solve. Conceptual reasoning — 10 seconds.

Q4

Easy

Answer: C) {2, 6, 10, 14, 18}

Step-by-Step Solution:

All sets have the same mean of 10.

Set A: {10,10,10,10,10} — $SD = 0$.

Set B: {8,9,10,11,12} — values spread 2 from mean.

Set C: {2,6,10,14,18} — values spread up to 8 from mean.

Set D: {9,10,10,10,11} — values spread only 1 from mean.

Set C has the greatest spread, so it has the greatest SD.

Why Other Options Are Wrong:

- A) $SD = 0$ — no spread at all.
- B) Moderate spread (deviations of 2, 1, 0, 1, 2).
- D) Very small spread (deviations of 1, 0, 0, 0, 1).

TRAP ALERT

Students might try to calculate each SD exactly. On the DSAT, visual comparison of spread is faster and sufficient.

KEY INSIGHT

SD measures spread from the mean. The set with values farthest from the mean has the largest SD.

FAST METHOD

Look for the set with the largest range or the values farthest from center. {2,6,10,14,18} wins easily.

VERDICT

Hand-solve. Visual comparison — 10 seconds.

Q5

Easy

Answer: B) The mean only

Step-by-Step Solution:

Adding a constant to every value shifts the entire distribution.

New mean = old mean + 5. So the mean changes.

Standard deviation measures spread, which is unchanged by a uniform shift.

New SD = old SD. The standard deviation does NOT change.

Why Other Options Are Wrong:

- A) SD does not change when a constant is added.
- C) Only the mean changes, not the SD.
- D) The mean definitely changes (increases by 5).

TRAP ALERT

This is one of the most tested SD properties. Adding a constant shifts the mean but NOT the SD. Multiplying by a constant changes BOTH.

KEY INSIGHT

Adding c to all values: mean increases by c , SD unchanged. Multiplying by c : mean multiplied by c , SD multiplied by $|c|$.

FAST METHOD

Add constant $\rightarrow$ only mean changes. Multiply $\rightarrow$ both change. Memorize this rule.

VERDICT

Hand-solve. Rule recall — 5 seconds.

Q6

Easy

Answer: B) Student B

Step-by-Step Solution:

Standard deviation depends on how far each value is from the mean.

The value 15 (Student B) is likely much farther from the mean than 2 (Student A).

Values farther from the mean contribute more to the standard deviation.

Why Other Options Are Wrong:

- A) Being small does not mean it contributes more — distance from mean matters.
- C) They are at different distances from the mean, so they contribute differently.
- D) Every value contributes to the SD calculation.

TRAP ALERT

The question tests whether you know that SD is about distance from the MEAN, not the size of the number itself.

KEY INSIGHT

Each value's contribution to SD is proportional to $(\text{value} - \text{mean})^2$. The farther from the mean, the more it contributes.

FAST METHOD

15 is more extreme (farther from the likely mean around 5-8), so Student B contributes more.

VERDICT

Hand-solve. Conceptual — 10 seconds.

Q7

Medium

Answer: A) Mean = 16, SD = 3

Step-by-Step Solution:

Each plant grows exactly 2 cm, so 2 is added to every value.

Adding a constant to all values: mean increases by 2 ($14 + 2 = 16$).

SD is unchanged because the spread does not change.

New mean = 16 cm, new SD = 3 cm.

Why Other Options Are Wrong:

- B) SD stays at 3, not 5.
- C) Mean changes to 16, not stays at 14.
- D) Neither value is 6.

TRAP ALERT

Identical growth for every plant is adding a constant — shifts the mean, not the spread.

KEY INSIGHT

Uniform shift = add constant = mean changes, SD stays.

FAST METHOD

Add 2 to mean: $14+2=16$. SD unchanged: 3. Done.

VERDICT

Hand-solve. Rule application — 5 seconds.

Q8

Medium

Answer: C) 2**Step-by-Step Solution:**

When every value is multiplied by a constant c , the new SD = $|c| \times$ original SD.

Here $c = 2$.

New SD = $2 \times$ original SD.

So the new SD is 2 times the original.

Why Other Options Are Wrong:

A) $1/2$ — inverts the multiplier.

B) 1 — assumes SD is unchanged (only true for addition, not multiplication).

D) 4 — squares the multiplier (that would be for variance, not SD).

TRAP ALERT

Option D (4 times) is the variance trap. Variance scales by $c^2 = 4$, but SD scales by $|c| = 2$.

KEY INSIGHT

SD scales linearly with multiplication: new SD = $|c| \times$ old SD. Variance scales quadratically: new variance = $c^2 \times$ old variance.

FAST METHOD

Multiply all values by 2 $\rightarrow$ SD doubles. Answer = 2.

VERDICT

Hand-solve. Rule recall — 5 seconds.

Q9

Medium

Answer: B) Group Y scores more spread out**Step-by-Step Solution:**

Both groups have the same mean (80), but Group Y has SD = 12 vs. Group X SD = 5.

Higher SD means greater spread from the mean.

Group Y scores are more spread out.

Why Other Options Are Wrong:

A) Median is not determined by SD alone.

C) Higher SD means Group Y (not X) has wider spread.

D) Different SDs mean different ranges are likely.

TRAP ALERT

Students confuse SD with range. They are related but not identical. However, higher SD reliably indicates more spread.

KEY INSIGHT

SD directly measures average distance from the mean. Higher SD = more spread. This is the definition.

FAST METHOD

Compare SDs: $12 > 5$. More spread. Done.

VERDICT

Hand-solve. Direct comparison — 5 seconds.

Q10

Medium

Answer: B) 2.4

Step-by-Step Solution:

$z\text{-score} = (\text{value} - \text{mean}) / \text{SD}$

$$z = (14 - 8) / 2.5 = 6 / 2.5 = 2.4$$

The value is 2.4 standard deviations above the mean.

Why Other Options Are Wrong:

A) 2 — rounds 2.4 down.

C) 3.5 — divides $14/2.5$ /something incorrectly.

D) 5.6 — computes $14/2.5 = 5.6$ (forgets to subtract the mean).

TRAP ALERT

Option D is the classic trap: dividing the raw value by SD instead of first subtracting the mean. Always compute (value - mean) first.

KEY INSIGHT

$z\text{-score} = (x - \text{mean}) / \text{SD}$. This formula tells you how many SDs a value is from the mean.

FAST METHOD

$$(14 - 8) / 2.5 = 6 / 2.5 = 2.4.$$

VERDICT

Hand-solve. One subtraction and one division — 10 seconds.

Q11

Medium

Answer: 3.5

Step-by-Step Solution:

Data: {5, 8, 10, 12, 15}. Mean = 10.

Deviations from mean: -5, -2, 0, 2, 5.

Squared deviations: 25, 4, 0, 4, 25.

Sum of squared deviations = $25 + 4 + 0 + 4 + 25 = 58$.

Population variance = $58/5 = 11.6$.

Population SD = $\sqrt{11.6} = 3.406$ (approx 3.4).

Rounded to nearest tenth: 3.4.

TRAP ALERT

Grid-in: the question specifies POPULATION SD (divide by n), not sample SD (divide by $n-1$). Using $n-1$ gives $\sqrt{58/4} = \sqrt{14.5} = 3.81$. Read carefully!

KEY INSIGHT

Population SD divides by n . Sample SD divides by $n-1$. The DSAT will specify which one, or the answer choices will make it clear.

FAST METHOD

Memorize the process: deviations $\rightarrow$ square $\rightarrow$ average $\rightarrow$ square root. With symmetric data, pair up deviations: two 5s, two 2s, one 0.

VERDICT

Desmos is faster. Enter the data into a list and use `stdev()` or `stdevp()` function to get the exact answer without manual computation.

Q12

Medium

Answer: C) 40

Step-by-Step Solution:

Within one SD of the mean means between mean - SD and mean + SD.

Range: $45 - 6 = 39$ to $45 + 6 = 51$.

Check each option: 33 (no), 38 (no), 40 (yes, $39 \leq 40 \leq 51$), 52 (no, $52 > 51$).

Answer: 40.

Why Other Options Are Wrong:

A) 33 is 2 SDs below the mean ($33 < 39$).

B) 38 is just outside the range ($38 < 39$).

D) 52 is just outside the range ($52 > 51$).

TRAP ALERT

Option B (38) is very close to the boundary but falls just outside. Precise boundary calculation matters!

KEY INSIGHT

Within 1 SD = [mean - SD, mean + SD]. Within 2 SDs = [mean - 2SD, mean + 2SD]. Know these intervals.

FAST METHOD

39 to 51. Only 40 falls in this range.

VERDICT

Hand-solve. Quick interval check — 10 seconds.

Q13

Medium

Answer: D) Same mean, Class 2 greater SD

Step-by-Step Solution:

Class 1 mean: $(70+72+74+76+78)/5 = 370/5 = 74$.

Class 2 mean: $(60+70+74+78+88)/5 = 370/5 = 74$.

Means are equal (both 74).

Class 1 values are tightly clustered (range = 8).

Class 2 values are widely spread (range = 28).

Class 2 has greater SD.

Why Other Options Are Wrong:

A) Means are equal, and Class 1 has SMALLER SD.

B) Correct about mean and SD relationship, but says "smaller" SD for Class 1 which is true — but the option says Class 1 has same mean, which is correct. However, D is more precise.

C) Means are equal, not different.

TRAP ALERT

Both classes intentionally have the same sum (370), so the same mean. The difference is entirely in spread. Students who do not compute both means will get this wrong.

KEY INSIGHT

Same mean does NOT mean same SD. Always evaluate spread separately from center.

FAST METHOD

Both sums = 370, so both means = 74. Compare ranges: 8 vs 28. Larger range -> larger SD.

VERDICT

Hand-solve. Sum comparison + range check — 20 seconds.

Q14

Medium

Answer: B) It decreases

Step-by-Step Solution:

The new employee's commute (35 min) equals the current mean.

Adding a value equal to the mean pulls no additional spread.

But the denominator (n) increases from 50 to 51, while the sum of squared deviations stays the same.

So variance = (same sum of squares) / (larger n) = smaller variance.

Therefore SD decreases (slightly).

Why Other Options Are Wrong:

A) Adding a value at the mean does not increase spread.

C) It does not stay exactly the same because n changes.

D) It can be determined — adding a value at the mean always decreases population SD.

TRAP ALERT

Students often think adding a value at the mean has "no effect." It does: the numerator of the variance formula stays the same but the denominator grows, so the SD decreases slightly.

KEY INSIGHT

Adding a data point equal to the mean always decreases the population SD (because the denominator increases while the numerator of squared deviations does not).

FAST METHOD

Value = mean => contributes 0 to sum of squares, but n increases => SD decreases.

VERDICT

Hand-solve. Conceptual reasoning — 15 seconds.

Q15

Medium

Answer: 1.73

Step-by-Step Solution:

Original data: {20, 24, 24, 28, 32, 36}. Sum = 164. Mean = $164/6 = 27.333\dots$

Remove 36: new data = {20, 24, 24, 28, 32}. Sum = 128. Mean = $128/5 = 25.6$.

Change in mean = $27.333\dots - 25.6 = 1.733\dots$

Rounded to nearest hundredth: 1.73.

TRAP ALERT

The question asks how much the mean **CHANGES** (the difference), not the new mean itself. Students who give 25.6 are answering the wrong question.

KEY INSIGHT

When removing a value above the mean, the mean decreases. The change = (removed value - old mean) $\times (1/(n-1))$ approximately.

FAST METHOD

Old mean = 27.33. New mean = 25.6. Difference = 1.73.

VERDICT

Desmos is helpful — enter the data as a list and compute mean() with and without the value. 15 seconds.

Q16

Medium

Answer: C) Both the mean and the SD

Step-by-Step Solution:

Original: {14,14,14,14,14,14}. Mean = 14, SD = 0.

Add 21: new set has 7 values. Mean = $(84+21)/7 = 105/7 = 15$.

Mean increased from 14 to 15.

SD was 0. Now there is variation (21 differs from the rest), so $SD > 0$.

Both mean and SD increase.

Why Other Options Are Wrong:

A) SD also increases (from 0 to a positive value).

B) Mean also increases (from 14 to 15).

D) Both definitely change.

TRAP ALERT

Since the original SD was 0, any new value that differs from the existing values will increase the SD. The mean also changes because 21 is not equal to 14.

KEY INSIGHT

When all values are identical: $SD = 0$ and mean = that value. Adding any different value changes both.

FAST METHOD

All 14s + one 21: mean shifts up, SD goes from 0 to positive. Both increase.

VERDICT

Hand-solve. Logical reasoning — 10 seconds.

Q17**Hard****Answer: B) 68%****Step-by-Step Solution:**

The distribution is approximately symmetric (approximately normal).

$$67 = 75 - 8 = \text{mean} - 1 \text{ SD.}$$

$$83 = 75 + 8 = \text{mean} + 1 \text{ SD.}$$

By the empirical rule (68-95-99.7), approximately 68% of data falls within 1 SD of the mean.

Why Other Options Are Wrong:

A) 50% would be for the range from the mean to somewhere specific, or for a different interval.

C) 95% applies to within 2 SDs of the mean.

D) 99.7% applies to within 3 SDs of the mean.

TRAP ALERT

The empirical rule applies to approximately normal distributions: 68% within 1 SD, 95% within 2 SDs, 99.7% within 3 SDs. Students must recognize that 67-83 spans exactly 1 SD on each side.

KEY INSIGHT

68-95-99.7 rule: memorize these three numbers. They apply whenever the distribution is approximately symmetric/normal.

FAST METHOD

67 to 83 = mean $\pm$ 1SD. Empirical rule $\rightarrow$ 68%.

VERDICT

Hand-solve. Rule recall — 5 seconds.

Q18**Hard****Answer: A) 3S****Step-by-Step Solution:**

Transformation: each value x becomes $3x + 7$.

Adding a constant (7) does NOT change the SD.

Multiplying by a constant (3) multiplies the SD by $|3| = 3$.

Order of operations: multiply first (SD becomes 3S), then add 7 (SD stays 3S).

New SD = 3S.

Why Other Options Are Wrong:

B) $3S + 7$ — addition does not affect SD.

C) 9S — squares the multiplier (this applies to variance, not SD).

D) $9S + 7$ — both errors combined.

TRAP ALERT

The biggest trap is option B ($3S + 7$): students think the added constant also adds to SD. It does not! Also, C uses the variance scaling (squared) instead of SD scaling.

KEY INSIGHT

For transformation $y = ax + b$: new mean = $a(\text{old mean}) + b$, new SD = $|a|(\text{old SD})$. The constant b has NO effect on SD.

FAST METHOD

Multiply by 3 $\rightarrow$ SD $\times$ 3. Add 7 $\rightarrow$ SD unchanged. Final = 3S.

VERDICT

Hand-solve. Transformation rule — 5 seconds.

Q19

Hard

Answer: C) SD of X > SD of Y

Step-by-Step Solution:

A wider IQR (larger box in boxplot) means the middle 50% of data is more spread out.

Greater spread in the middle 50% strongly suggests greater overall spread.

Therefore SD of X is most likely greater than SD of Y.

Why Other Options Are Wrong:

A) A wider box means more spread, not less.

B) Different IQRs suggest different SDs.

D) While SD is not perfectly determined by boxplots, a much larger IQR makes it very likely that $SD(X) > SD(Y)$.

TRAP ALERT

Option D is technically somewhat valid, but the word "most likely" in the correct answer makes C the best choice. The DSAT uses "most likely" to allow for probabilistic reasoning.

KEY INSIGHT

IQR and SD both measure spread. A much larger IQR almost always means a larger SD.

FAST METHOD

Bigger box = more spread = higher SD. X wins.

VERDICT

Hand-solve. Visual reasoning from boxplot — 10 seconds.

Q20

Hard

Answer: 3.7

Step-by-Step Solution:

Original: $n=6$, $\text{mean}=12$, $\text{SD}=4$. Variance = 16.

Sum of squared deviations = $n \times \text{variance} = 6 \times 16 = 96$.

Add 7th value = 12 (equals the mean, contributes 0 to sum of squared deviations).

New sum of squared deviations = $96 + 0 = 96$.

New mean = $(6 \times 12 + 12)/7 = 84/7 = 12$ (unchanged).

New population variance = $96/7 = 13.714\dots$

New population SD = $\sqrt{96/7} = \sqrt{13.714} = 3.703\dots$

Rounded to nearest tenth: 3.7.

TRAP ALERT

Grid-in: students might think adding a value equal to the mean leaves SD unchanged. But the population SD formula divides by n , and n increased from 6 to 7, reducing the variance.

KEY INSIGHT

New population SD = $\sqrt{\text{old sum of squares} / \text{new } n}$. The sum of squares does not change, but n increases.

FAST METHOD

Sum of squares = $6 \times 16 = 96$. New SD = $\sqrt{96/7} = \sqrt{13.71} = 3.7$.

VERDICT

Desmos is helpful — calculate $\sqrt{96/7}$ directly. 15 seconds.

Q21

Hard

Answer: A) Machine A

Step-by-Step Solution:

Machine A: mean = 500, SD = 2. The range 496-504 is mean $\pm$ 2 SDs.

By the empirical rule, $\sim 95\%$ of Machine A bottles are in this range.

Machine B: mean = 500, SD = 8. The range 496-504 is mean $\pm$ 0.5 SDs.

Only about 38% of Machine B bottles are in this range.

Machine A has far fewer rejected bottles.

Why Other Options Are Wrong:

B) Machine B has larger SD, so more bottles fall outside the range.

C) Different SDs mean different rejection rates.

D) The information is sufficient to determine the answer.

TRAP ALERT

Students need to convert the absolute range (496-504) into number of SDs for each machine. For Machine A it is 2 SDs (covers 95%), for Machine B it is only 0.5 SDs (covers $\sim 38\%$).

KEY INSIGHT

Smaller SD = more consistent = fewer rejections when the acceptable range is fixed.

FAST METHOD

Machine A: range covers ± 2 SD ($\sim 95\%$ pass). Machine B: range covers ± 0.5 SD ($\sim 38\%$ pass). A wins.

VERDICT

Hand-solve. Convert range to SDs and compare — 15 seconds.

Q22

Hard

Answer: A) 73

Step-by-Step Solution:

z-score formula: $z = (x - \text{mean}) / \text{SD}$

$$-1.5 = (x - 82) / 6$$

$$-1.5 \times 6 = x - 82$$

$$-9 = x - 82$$

$$x = 82 - 9 = 73$$

Why Other Options Are Wrong:

B) 75 — uses $z = -1.0$ or adds incorrectly.

C) 79 — uses $z = -0.5$.

D) 91 — uses $z = +1.5$ (wrong sign).

TRAP ALERT

The negative z -score means the student scored BELOW the mean. Option D uses a positive z -score, which would place the student above the mean.

KEY INSIGHT

$z = (x - \text{mean})/\text{SD}$. Solve for x : $x = \text{mean} + z(\text{SD})$. Negative z means below the mean.

FAST METHOD

$$x = 82 + (-1.5)(6) = 82 - 9 = 73.$$

VERDICT

Hand-solve. Plug into formula — 10 seconds.

Q23**Hard****Answer: 2.83**

Step-by-Step Solution:

Data: {72, 74, 76, 78, 80}. Mean = $380/5 = 76$.

Deviations: -4, -2, 0, 2, 4.

Squared deviations: 16, 4, 0, 4, 16.

Sum = 40.

Population variance = $40/5 = 8$.

Population SD = $\sqrt{8} = 2.828\dots$

Rounded to nearest hundredth: 2.83.

TRAP ALERT

Population SD (divide by $n = 5$). If using sample SD (divide by $n-1 = 4$), you get $\sqrt{10} = 3.16$, which is wrong for this question.

KEY INSIGHT

For evenly spaced data centered at the mean, the deviations are symmetric. This makes the calculation clean.

FAST METHOD

Symmetric deviations: two 4s, two 2s, one 0. Sum of squares = $2(16) + 2(4) = 40$. SD = $\sqrt{40/5} = \sqrt{8} = 2.83$.

VERDICT

Desmos is slightly faster — enter the list and use `stdevp()`. But hand calculation is straightforward with symmetric data.

Q24**Hard****Answer: A) $3a + 2b = 100$ and variance = 36****Step-by-Step Solution:**

Mean = $(3a + 2b)/5 = 20$, so $3a + 2b = 100$.

Population variance = $SD^2 = 6^2 = 36$.

Variance = $[3(a-20)^2 + 2(b-20)^2] / 5 = 36$.

This matches option A.

Why Other Options Are Wrong:

B) Sets variance = 6 instead of 36 (forgets to square the SD).

C) Uses $3a + 2b = 20$ (forgets to multiply mean equation by 5).

D) Treats frequencies as 1 each instead of 3 and 2.

TRAP ALERT

$SD = 6$ but the variance formula uses $SD^2 = 36$. Option B uses 6 instead of 36 — a very common trap.

KEY INSIGHT

Always square the SD when using the variance formula. $SD^2 = \text{variance}$.

FAST METHOD

Mean equation: $3a + 2b = 100$. Variance equation with $SD^2 = 36$. Match to option A.

VERDICT

Hand-solve. Setup question — no actual solving needed. 20 seconds.

Q25**Hard****Answer: B) $2d$** **Step-by-Step Solution:**

Data set 2: $\{k-2, k-1, k, k+1, k+2\}$ has $SD = d$.

New data set: multiply each value by 2: $\{2(k-2), 2(k-1), 2k, 2(k+1), 2(k+2)\}$.

This is equivalent to multiplying every value in Data set 2 by 2.

Multiplying all values by a constant c scales SD by $|c|$.

New $SD = 2d$.

Why Other Options Are Wrong:

A) d — assumes multiplication does not affect SD (only true for addition).

C) $4d$ — uses variance scaling (c^2) instead of SD scaling ($|c|$).

D) $d + 2k$ — incorrectly adds a term related to the mean.

TRAP ALERT

Option C ($4d$) is the variance trap again. Variance scales by c^2 , but SD scales by $|c|$. Also note that the value of k is irrelevant — it only affects the mean, not the SD.

KEY INSIGHT

The SD of Data set 2 does not depend on k at all (shifting by k does not affect spread). Multiplying by 2 doubles the SD.

FAST METHOD

Multiply by 2 $\rightarrow$ SD doubles. k is irrelevant. Answer = $2d$.

VERDICT

Hand-solve. Transformation rule — 5 seconds.

TOPIC SUMMARY

Core Concept

Standard deviation (SD) measures how spread out data values are from the mean. It is the square root of the variance. The DSAT tests conceptual understanding far more than calculation — you rarely need to compute SD from scratch.

Key Rules to Memorize

SD = 0 if and only if all values are identical.

Adding a constant c to all values: mean shifts by c , SD unchanged.

Multiplying all values by c : mean scales by c , SD scales by $|c|$.

Variance = SD squared. (Do not confuse SD and variance in formulas.)

$z\text{-score} = (\text{value} - \text{mean}) / \text{SD}$.

The Empirical Rule (68-95-99.7)

For approximately normal distributions: 68% within 1 SD, 95% within 2 SDs, 99.7% within 3 SDs.

Most Common DSAT Question Types

1. Compare SDs of two data sets (no calculation — judge by spread).
2. Effect of adding/multiplying a constant on mean and SD.
3. Effect of adding/removing a data point on SD.
4. z -score calculations.
5. Empirical rule (percentage within k SDs).

Top 3 Traps to Avoid

1. Confusing SD with variance (forgetting to square or square-root).
2. Thinking adding a constant affects SD (it does not).
3. Using variance scaling (c -squared) when the question asks about SD (use $|c|$).

Desmos vs. Hand-Solve Summary

80% of SD questions are conceptual and fastest by hand. Use Desmos only for actual SD computation (entering data and using `stdevp()` or `stdev()`).